A very large drone (Source: www.hyclight.com)

collateral damage. Their civil applications include policing, firefighting, inspection of power lines and pipelines, delivering packages, etc.

As drones are becoming increasingly versatile, stealthy, and convenient airborne weapons, there is need for hostile drones to be detected quickly, identified, localised, and neutralised. Counter-drone systems, also called counter-unmanned aerial systems (C-UAS), are used to detect and neutralise the hostile unmanned aerial systems while in flight to protect areas such as critical infrastructure, airports, large public spaces, and military installations.

The evolution of drones

Drones were initially used for military applications as the defence forces were quick to recognise their benefit for wartime strategies and tactical usage. Later, the use expanded to commercial, scientific, agricultural, research, and other applications.

Drones were initially developed as balloons and aerial targets for training, target practice, bomb detection, and surveillance. By the early 1900s, the US military were using them as practice aerial targets.

In 1915, Nikola Tesla suggested their use as unmanned aerial combat vehicles and a self-propelled drone was used as an aerial target in 1916.

A US Air Force unmanned combat aircraft used during the Vietnam war (Source: https://aviationoiloutlet.com)

During World War II both Allied and German forces used drones as targets to train aircraft gunners and aid-in missions.

However, not much development took place until the Vietnam War. From the early years of the Vietnam War, the US Air Force started using the unmanned combat aircrafts to cut down on pilot casualties. The development got speeded up for decoy and surveillance applications when an American spy plane was shot down by the Soviet Union in 1960.

During the 1990s, the US began developing more advanced and cost-efficient drones. Today, they have become an essential part of military missions and operations and are in high demand by nearly every military and law-enforcement agency.

Drones' use for non-military purposes started in 2006, when the Federal Aviation Administration of the USA issued the first commercial drone permit. Government agencies quickly began using drones for disaster relief and border surveillance while corporates began using them for commercial applications like pipeline inspection, crop evaluation, and security.

In 2013, Amazon started using drones for delivering orders. In 2018, companies around the world began developing drones for taxi services, photography, and indoor applications.

Drones have a bright future with their increasing use by the industry and applications into many more fields. Negative forces like terrorists and drug cartels are at work too, and are increasingly using drones for smuggling contrabands, dropping arms, explosives and IEDs, and terroristic raids. It is estimated that currently over three-dozen countries and multiple terrorist groups and non-state actors possess weaponised drones in their arsenal.

Counter-drone systems identify and neutralise the unknown drones by taking into account their flight path, physical properties such as size and kinetic energy, the number of swarming drones, and the likely damage which could be caused. Counter-drone systems are preferably portable and have the capability of defending the facilities like military establishments and sites, airports, sports stadia, and outdoor/indoor convention sites by quickly detecting multiple drones at sufficient distances.

Counter-drone systems have two parts—monitoring equipment and countermeasure equipment.

Drone monitoring equipment

Drone monitoring equipment detect the approaching drones, classify them from other types of objects—like planes, birds, trains, and automobiles, identify if they are hostile or illegal, locate and track their flight path, determine the hazard level, and